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BANGARAM

STUDENT ENROLLMENT AND COURSE MANAGEMENT SYSTEM

Submitted

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ABSTRACT

The Student Enrollment and Course Management System is a web application developed using ASP.NET and C#.

This system helps educational institutions manage student details, course information, and enrollment processes efficiently using SQL Server database.

The system allows administrators to add students, create courses, enroll students, and retrieve information quickly.

It reduces manual work and improves academic management efficiency.

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INTRODUCTION

Educational institutions maintain large amounts of student and course information. Managing this manually can lead to errors and inefficiency.

The Student Enrollment and Course Management System automates the management of students, courses, and enrollments using ASP.NET MVC and SQL Server.

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OBJECTIVES

- To maintain student records
- To manage course details
- To allow student enrollment into courses
- To provide a centralized database system
- To reduce manual paperwork

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EXISTING SYSTEM

In the existing system, student and course data are recorded manually.

Problems include time consumption, higher chance of errors, difficulty searching records, and data redundancy.

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PROPOSED SYSTEM

The proposed system is a web-based application that stores student and course data digitally.

Features include student registration, course creation, enrollment, and fast data retrieval.

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SYSTEM REQUIREMENTS

Hardware: Computer with 4GB RAM, 500GB Hard Disk.

Software: Visual Studio, ASP.NET MVC, SQL Server, Windows OS.

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SYSTEM ARCHITECTURE

The system follows three-tier architecture: Presentation Layer, Business Logic Layer, and Data Layer.

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DATABASE DESIGN

The system uses SQL Server to store student, course, and enrollment data.

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STUDENT TABLE (SQL)

```
CREATE TABLE Students(
StudentId INT PRIMARY KEY IDENTITY,
Name VARCHAR(100),
RollNumber VARCHAR(20)
);
```

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COURSE TABLE (SQL)

```
CREATE TABLE Courses(
CourseId INT PRIMARY KEY IDENTITY,
CourseName VARCHAR(100),
Credits INT
);
```

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ENROLLMENT TABLE (SQL)

```
CREATE TABLE Enrollments(
EnrollmentId INT PRIMARY KEY IDENTITY,
StudentId INT,
CourseId INT
);
```

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ER DIAGRAM

Student → Enrollment → Course

Student: StudentId, Name, RollNumber

Course: CourseId, CourseName, Credits

Enrollment: EnrollmentId, StudentId, CourseId

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DATA FLOW DIAGRAM

Admin → Student Enrollment System → Database

Functions: Add Student, Add Course, Enroll Student, View Records

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ASP.NET MODEL CLASSES

```
public class Student
{
public int StudentId { get; set; }
public string Name { get; set; }
public string RollNumber { get; set; }
}
```

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CONTROLLER CODE

```
public class StudentController : Controller
{
ApplicationDbContext db = new ApplicationDbContext();

public ActionResult Index()
{
return View(db.Students.ToList());
}
}
```

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SAMPLE OUTPUT

Student List

Arsheen G | 23133080500122005

Bakthinayagi M | 23133080500122007

Santhiya B | 23133080500122038

Sabina S | 23133080500122033

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ADVANTAGES

- Easy to manage student data
- Fast data retrieval
- Reduces paperwork
- Secure database storage

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TESTING

Testing methods include Unit Testing, Integration Testing, and System Testing.

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CONCLUSION

The Student Enrollment and Course Management System simplifies the management of student and course data. It improves efficiency and ensures accurate storage using ASP.NET and SQL Server.

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Student Enrollment and Course Management

Admin

Home Students Courses Enrollments

Students

Add New Student

Name	Roll Number
Arsheen G	23133080500122005
Bakthinayagi M	23133080500122007
Santhiya B	23133080500122038
Sabina S	23133080500122033

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Student Enrollment and Course Management

Admin

Home Students Courses Enrollments

Courses

Add New Course

Course Name	Credits
Database Systems	4
Web Development	3
Software Engineering	4

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Student Enrollment and Course Management

Admin

Home Students Courses Enrollments

Enrollments

Enroll New Student

Student Name	Course Name	Credits
Arsheen G	Database Systems	4
Bakthinayagi M	Web Development	3
Santhiya B	Software Engineering	4
Sabina S	Database Systems	4

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Student Enrollment and Course Management

Admin

Home Students Courses Enrollments

Student List

Add New Course

Student Name	Course Name	Credits
Arsheen G	Database Systems	4
Bakthinayagi M	Web Development	3
Santhiya B	Software Engineering	4
Sabina S	Database Systems	4

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Student Enrollment and Course Management

Admin

Home Students Courses Enrollments

Enroll New Student

The main objectives of this project are:

- **Student**

Sabina S

Course:

Web Development

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Student Enrollment and Course Management

Admin

Home Students Courses Enrollments

Enroll New Student

Student:

Sabina S

Course:

Web Development

Enroll

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